7 Memory Units

Introduction

Memory is a device or circuit that is used to store the digital information for digital processing system. Earlier, the memories used were mostly of magnetic type but nowadays we use semiconductor memories for data storage. There are various types of semiconductor memories available, depending upon the size, cost and speed.

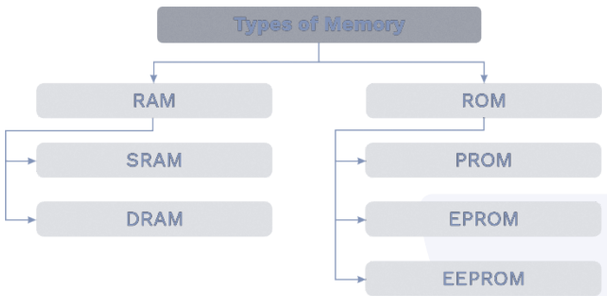


Fig. 7.1

Read Only Memory (ROM)

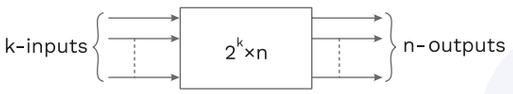


Fig. 7.2

- ROM has k-input lines and n-output lines. In general, 2k×n ROM will have internally one k×2k size decoder and n-OR gates.
- In ROM, input lines are called address lines and output lines are called data lines.
- ROM contains fixed AND array and programmable OR array.

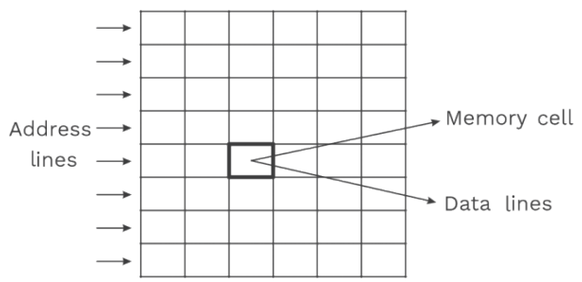


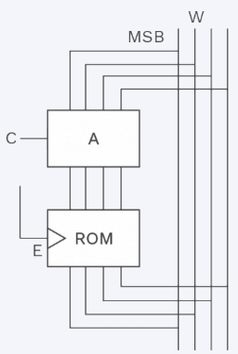
Fig. 7.3

- If we select the memory cell using address line, the information present inside the cell is given by the corresponding data line.

Previous Years' Questions

In the circuit shown below, A is a parallel-in parallel-out 4 bit register, which loads at the rising edge of the clock C. The input lines are connected to a 4 bit bus W. Its output acts as the input to a 16×4 ROM whose output is floating when the enable input E is 0.

Address	Data
0	0011
2	1111
4	0100
6	1010
8	1011
10	1000
12	0010
14	1000



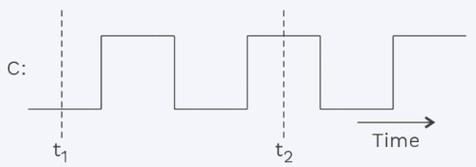


Fig. 7.4

The clock to the register is shown, and the data on the W bus at time t1 is 0110. The data on the bus at time t2 is

(A) 1111 (B) 1011
(C) 1000 (D) 0010

Solution: (C)

Specifications of ROM:

- ROM is read-only memory. So, we can only read the data but cannot change the data or write anything inside ROM.
- ROM is a non-volatile memory. It permanently stores the data.
- ROM is used to implement the combinational circuits.

Programmable Read-Only Memory (PROM)

- This type of memory is not programmed during the manufacturing process but contains all 1's (or all 0's) in every bit of the stored words.
- Some type of link or fuse is used. A broken link represents 1's state and an unbroken link represents the 0's state.
- The fuse or link in the PROM is broken by using the current pulses at the output terminal.
- An address is selected by the switch setting on the address lines and a pulse is applied to those output lines corresponding to locations where 0's are to be stored.
- These pulses will blow the fuse links, thereby creating the desired bit pattern. The process is then repeated for the next location.
- Special units called PROM programmers automatically carry out the sequence for programming.
- Here the interesting thing to note is that all procedures for programming ROMs are hardware procedures but the word "programming" is used.
- Because of this hardware procedure of ROM or PROM once programmed cannot be altered or programmed again.

Erasable Programmable Read-Only Memory (EPROM)

- The benefit of EPROM is that it can be restructured again even though it has been programmed previously.
- This type of memory uses an NMOS array with an isolated gate structure. The isolated gate has no electrical connections and stores electrical charge for indefinite period of time.
- The presence of stored charge represents logic 1 and absence of stored charge represents a logic 0.
- Erasure of a data bit is done by removing the gate charge.
- When an ultraviolet light is passed through the EPROM for a short period of time, the radiation discharges the internal gates that works as contacts.
- After erasure, the EPROM returns to its initial state and it can be again programmed.

Electrically erasable and programmable read-only memory (EEPROM)

It is an another type of reprogrammable memory where erasing is done electrically. Ultraviolet radiations are not used. The advantage is that the device can be erased without removing it from the socket. It is referred to as EEPROM or electrically alterable ROM (EAROM).

Example 1:

Implement the following functions using ROM-

$$F1 (A, B, C) = \sum m(0, 1, 2, 5, 7)$$

$$F2 (A, B, C) = \sum m(1, 4, 6)$$

Solution:

Here, output lines (data lines) = 2

Input lines (address lines) = 3

So, ROM size = 23 × 2

Decoder size = One 3×8 decoder

Number of OR-gate required = 2 OR-gates

Implementation of given function,

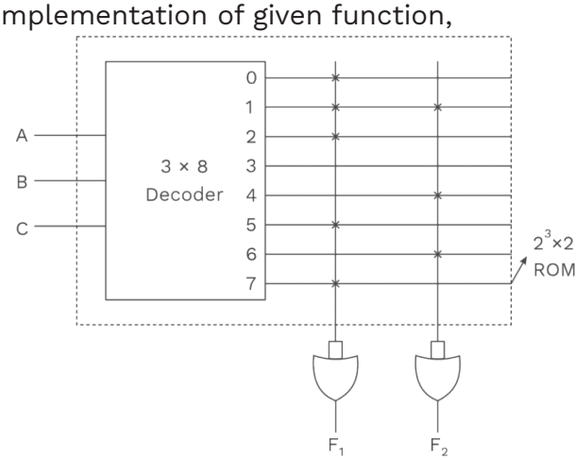


Fig. 7.5

Given,

$$F1 = Sm(0, 1, 2, 5, 7)$$

K-map simplification,

Previous Years' Questions

A 2 × 2 ROM array is built with the help of diodes as shown in the circuit below. Here W0 and W1 are signals that select the word lines and B0 and B1 are signals that are output of the sense amps based on the stored data corresponding to the bit lines during the read operation.

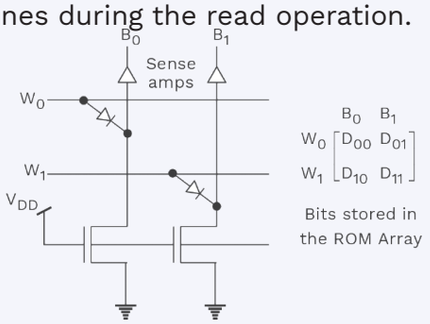


Fig. 7.6

During the read operations, the selected word line goes high and the other word line is in a high impedance state. As per the implementation shown in the circuit diagram above, what are the bits corresponding to Dij (where i = 0 or 1 and j = 0 or 1) stored in the ROM?

(A) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

Solution: (A)

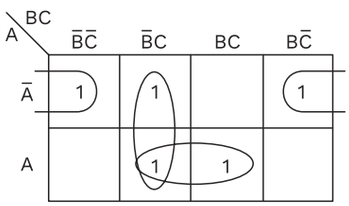


Fig. 7.7

$F_1 = AC + \bar{A}\bar{C} + \bar{B}C$

and,

$F_2 = \Sigma m(1, 4, 6)$

K-map simplification,

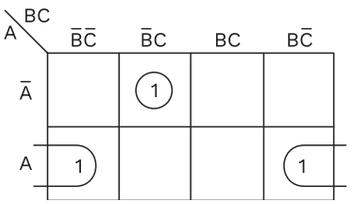


Fig. 7.8

$F_2 = A\bar{C} + \bar{A}\bar{B}C$

So, $F_1 = \bar{A}\bar{C} + \bar{B}C + AC$

$F_2 = A\bar{C} + \bar{A}\bar{B}C$

Implementation of given functions using diodes,

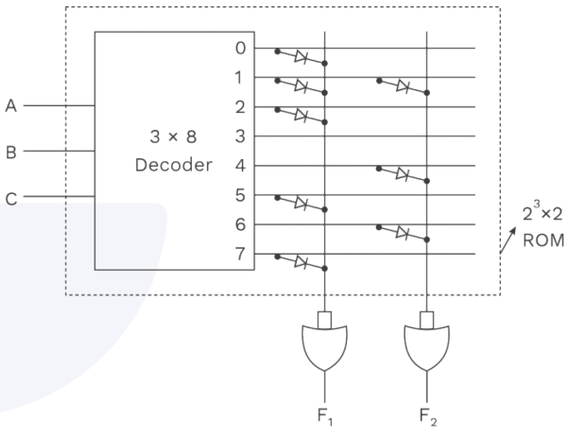


Fig. 7.9

So, $F_1 = \bar{A}\bar{C} + \bar{B}C + AC$

$F_2 = A\bar{C} + \bar{A}\bar{B}C$

Example 2:

Realize the following functions using a ROM.

F1 (A, B, C) = ∑m (0, 4, 7)

F2 (A, B, C) = ∑m (1, 3, 6)

F3 (A, B, C) = ∑m (1, 2, 4, 6)

Solution:

Here, output lines (data lines) = 3

Input lines (address lines) = 3

So, ROM size = 23 × 3

Decoder size = 3 × 8 decoder

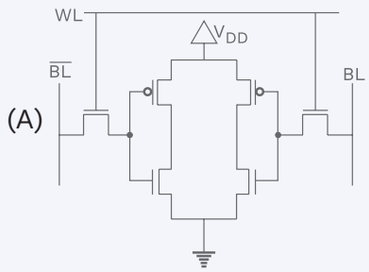
Number of OR-gate required = 3

Implementation of given function,

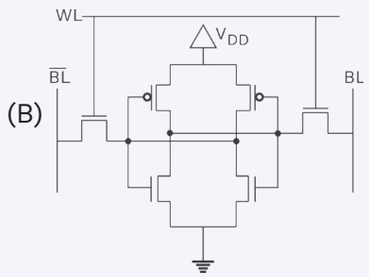
Previous Years' Questions

If WL is the word line and BL the bit line, an SRAM cell is shown in:

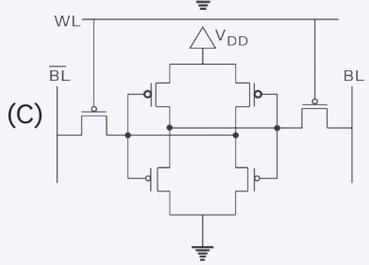
(A)



(B)



(C)



(D)

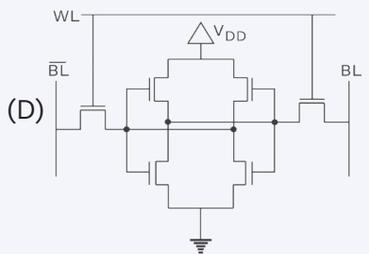


Fig. 7.10

Solution: (B)

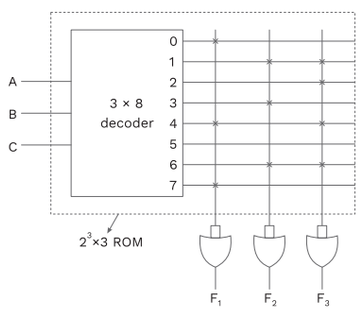


Fig. 7.11

So, $F_1 = \overline{A}\overline{B}\overline{C} + A\overline{B}\overline{C} + ABC$

$F_1 = \overline{B}\overline{C} + ABC$

$F_2 = \overline{A}\overline{B}C + \overline{A}B\overline{C} + AB\overline{C}$

$F_2 = \overline{A}C + AB\overline{C}$

$F_3 = \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + AB\overline{C}$

$F_3 = \overline{A}\overline{B}C + A\overline{C} + B\overline{C}$

Programmable Logic Array (PLA)

- PLA contains programmable AND array and programmable OR array.
- In PLA, product terms are generated using AND array. So, decoder is not required.

Example 3:

Implement the following functions using PLA.

$F1(A, B, C) = \sum m(0, 1, 3, 4)$

$F2(A, B, C) = \sum m(1, 2, 3, 4, 5)$

Solution:

Given,

$F1 = \sum m(0, 1, 3, 4)$

$F_1 = \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}BC + A\overline{B}\overline{C}$

$F_1 = \overline{B}\overline{C}(A + \overline{A}) + \overline{A}C(\overline{B} + B)$

$F_1 = \overline{B}\overline{C} + \overline{A}C$

and $F2 = \sum m(1, 2, 3, 4, 5)$

$F_2 = \overline{A}\overline{B}C + \overline{A}B\overline{C} + \overline{A}BC + A\overline{B}\overline{C} + A\overline{B}C$

$F_2 = \overline{A}B + \overline{A}C + A\overline{B}$

Implementation using PLA (Programmable AND and Programmable OR-array)

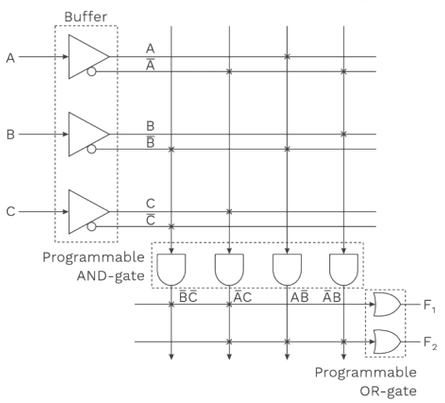


Fig. 7.12

So, $F_1 = \overline{B}\overline{C} + \overline{A}C$

$F_2 = \overline{A}B + \overline{A}C + A\overline{B}$

Previous Years' Questions

In a DRAM:

(A) Periodic refreshing is not required.

(B) Information is stored in a capacitor.

(C) Information is stored in a latch.

(D) Both read and write operations can be performed simultaneously.

Solution: (B)

Example 4:

Find the minterms of the given circuit (F1 and F2).

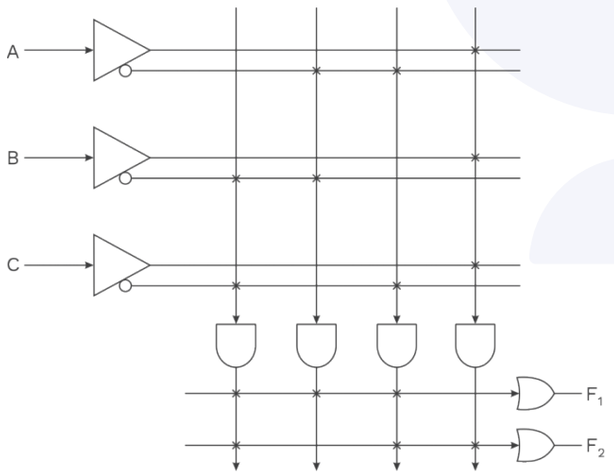


Fig. 7.13

Grey Matter Alert!

SRAM is a volatile memory, i.e., when power is turned OFF, all contents are destroyed.

Solution:

Given circuit,

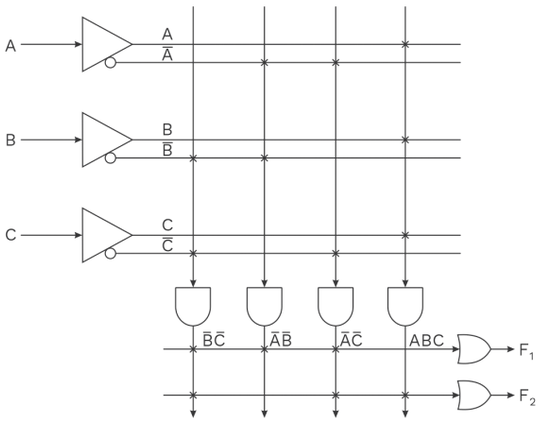


Fig. 7.14

So, $F_1 = \overline{B}\overline{C} + \overline{A}\overline{B} + \overline{A}\overline{C}$

A \ BC	$\overline{B}\overline{C}$	$\overline{B}C$	BC	$B\overline{C}$
$\overline{A}$	1	1		1
A	1			

F1 = ∑m (0, 1, 2, 4)

and $F_2 = \overline{B}\overline{C} + \overline{A}\overline{C} + ABC$

A \ BC	$\overline{B}\overline{C}$	$\overline{B}C$	BC	$B\overline{C}$
$\overline{A}$	1			1
A	1		1	

F2 = ∑m (0, 2, 4, 7)

Therefore, minterms are,

F1 = ∑m (0, 1, 2, 4)

F2 = ∑m (0, 2, 4, 7)

Grey Matter Alert!

SRAM has access time in order of 5–50 ns.

Programmable Array Logic (PAL)

- PAL contains programmable AND array and fixed OR array.
- In PAL, product terms are generated by using AND array. So, decoder is not required.

Example 5:

Design a PAL circuit for the given Boolean expressions.

$w(A,B,C,D) = \Sigma m(2,12,13)$

$x(A,B,C,D) = \Sigma m(7,8,9,10,11,12,13,14,15)$

$y(A,B,C,D) = \Sigma m(0,2,3,4,5,6,7,8,10,11,15)$

$z(A,B,C,D) = \Sigma m(1,2,8,12,13)$

Solution:

Step 1: First, we make K-maps and simplify as shown below.

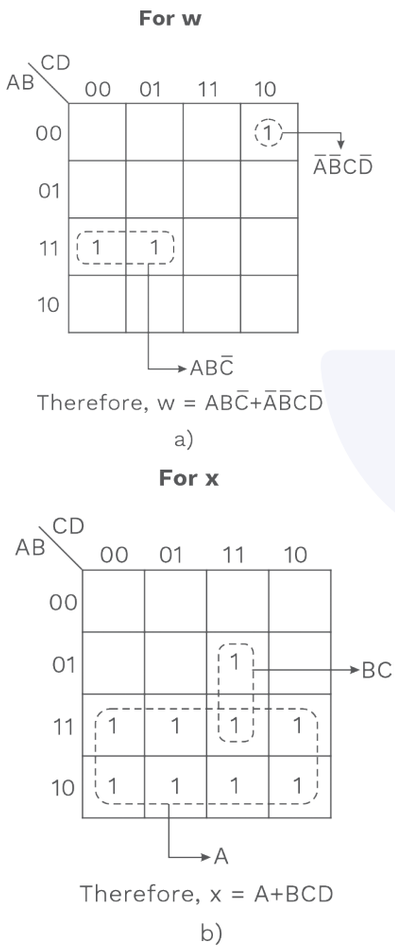


Fig. 7.15

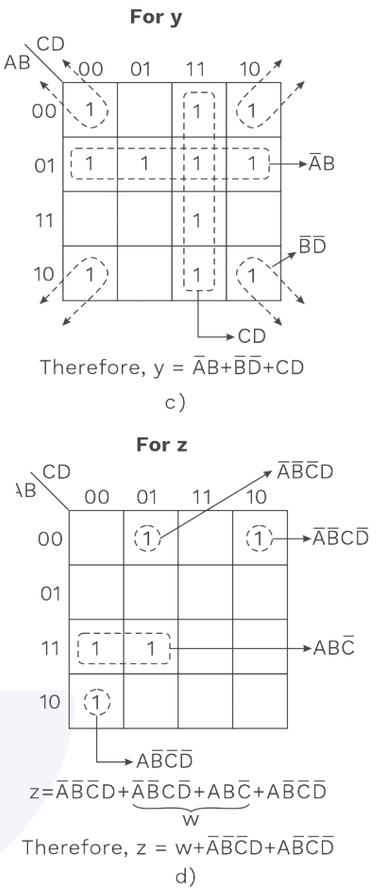


Fig. 7.16

We have to use a 4 input, 3 wire AND-OR structured PAL. Therefore, we must restrict the number of product terms in expression for each output to at the most 3. But z consists of 4-product terms. Hence, as shown in figure (d), we have reduced them to 3 by substituting $w = \bar{A}\bar{B}C\bar{D} + AB\bar{C}$. The outputs w and x contain only two product terms each. Hence, one AND gate for each one of them will remain unused. Then, we have to leave all the fuse links intact for an unused AND gate.

Step 2: Next, we prepare the PAL programming table as shown in table.

Product term No.	Product term	AND inputs					Outputs
		A	B	C	D	w	
1	$AB\bar{C}$	1	1	0	–	–	$w = AB\bar{C} + \bar{A}\bar{B}C\bar{D}$
2	$\bar{A}\bar{B}C\bar{D}$	0	0	1	0	–	
3	–	–	–	–	–	–	

Product term No.	Product term	AND inputs					Outputs
		A	B	C	D	w	
4	A	1	–	–	–	–	$x = A + BCD$
5	BCD	–	1	1	1	–	
6	–	–	–	–	–	–	
7	$\overline{A}B$	0	1	–	–	–	$y = \overline{A}B + \overline{B}\overline{D} + CD$
8	$\overline{B}\overline{D}$	–	0	–	0	–	
9	CD	–	–	1	1	–	
10	w	–	–	–	–	1	$z = w + \overline{A}\overline{B}\overline{C}D + A\overline{B}\overline{C}\overline{D}$
11	$\overline{A}\overline{B}\overline{C}D$	0	0	0	1	–	
12	$A\overline{B}\overline{C}\overline{D}$	1	0	0	0	–	

Table 7.1 PAL Programming Table

Step 3: Lastly, we carry out the implementation as shown below.

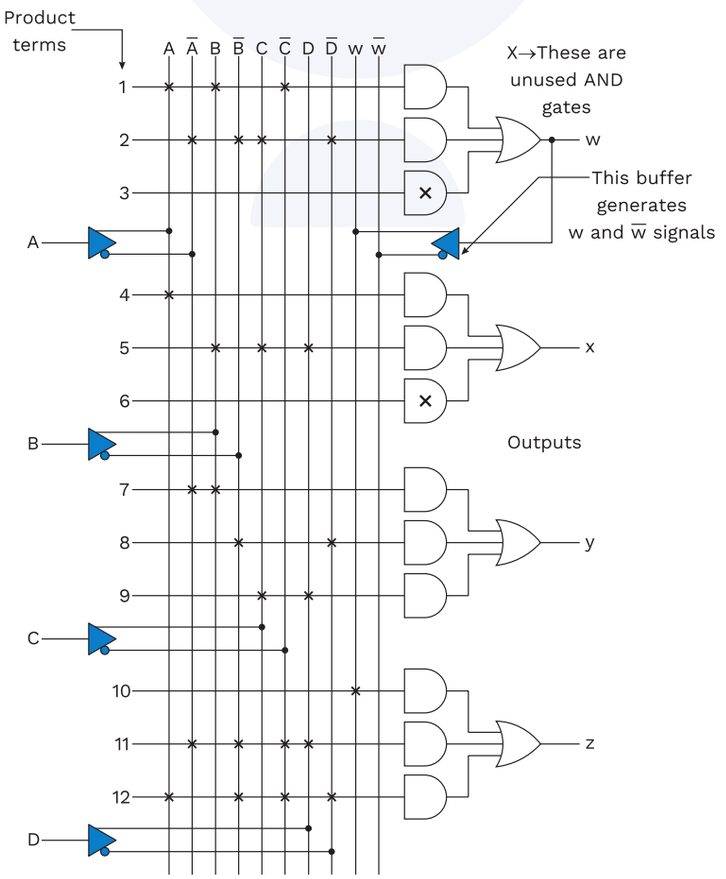


Fig. 7.17 Implementation using PAL

Example 6:

Draw a PAL circuit for the given truth table.

Inputs			Outputs			
x	y	z	A	B	C	D
0	0	0	0	1	0	0
0	0	1	1	1	1	1
0	1	0	1	0	1	1
0	1	1	0	1	0	1
1	0	0	1	0	1	0
1	0	1	0	0	0	1
1	1	0	1	1	1	0
1	1	1	0	1	1	1

Table. 7.2

Solution:

Step 1: First, we obtain simplified Boolean expression using 3-variable K-map as shown below:

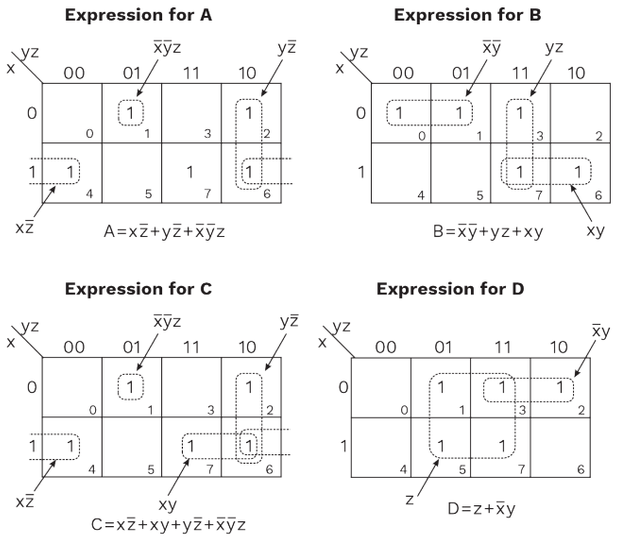


Fig. 7.18

The simplified expressions are:

$A = x\bar{z} + y\bar{z} + \bar{x}\,\bar{y}z$

$B = \bar{x}\,\bar{y} + xy + yz$

$C = x\bar{z} + y\bar{z} + \bar{x}\bar{y}z + xy$

$C = A + xy \quad [\because x\bar{z} + y\bar{z} + \bar{x}\,\bar{y}z = A]$

$D = z + \bar{x}y$

Step 2: PAL programming table is now redrawn.

Product Term No.	Product Term	AND Inputs				Outputs
		x	y	z	A	
1	$x\bar{z}$	1	–	0	–	A (sum of product 1,2,3)
2	$y\bar{z}$	–	1	0	–	
3	$\bar{x}\,\bar{y}z$	0	0	1	–	
4	$\bar{x}\,\bar{y}$	0	0	–	–	B (sum of product 4,5,6)
5	xy	1	1	–	–	
6	yz	–	1	1	–	
7	A	–	–	–	1	C (sum of product 7,8)
8	xy	1	1	–	–	
9	z	–	–	1	–	D (sum of product 9,10)
10	$\bar{x}\,y$	0	1	–	–	

Table. 7.3

Step 3: Logic diagram is shown below.

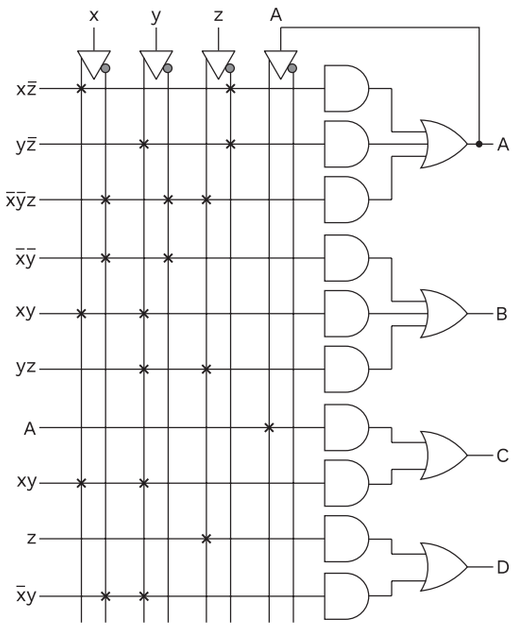


Fig. 7.19

Random Access Memory (RAM)

1. In RAM, we can read as well as change the stored data.
2. RAM stores the data at physical locations such as hard disks, CD-RWs, DVD-RWs and magnetic tapes.
3. There are two types of RAM:
   a) SRAM (Static RAM)
   b) DRAM (Dynamic RAM)

Previous Years' Questions

The minimum number of MOS transistors required to make a dynamic RAM cell is

(A) 1 (B) 2
(C) 3 (D) 4

Solution: (B)

Dram (Dynamic RAM)

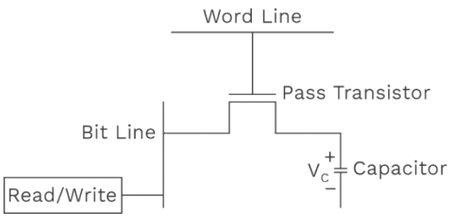


Fig. 7.20

- Memory is a combination of cells and each memory cell contains a DRAM cell.
- In the DRAM cell, the word line (WL) is also called address line and bit line (BL) is also called data line.

Read operation

- In DRAM, data is stored in the form of voltage across the capacitor (VC).
- Select the desired cell location using word line (WL) by applying logic '1' of the particular cell.
- If Read/Write (R / $\overline{W}$) and word line (WL) is logic '1', then the bit line (BL) acts as an output line and pass transistor acts as a short circuit (SC). So, at the data line (bit lines) capacitor voltage (VC) will store and read operation takes place in DRAM.

Grey Matter Alert!

The package density of SRAM to DRAM = 1:4.

Write operation

- Select the desired cell location using word line (WL) by applying logic-1 of the particular cell.
- If Read/Write (R / $\overline{W}$) is logic '0' and word line (WL) is logic '1', then bit line (BL) act as an input line and pass transistor act as a short circuit (SC). So, at the data line (bit line) capacitor voltage (VC) will store and the write operation takes place in DRAM.

Previous Years' Questions

A dynamic RAM consists of:
(A) 6 transistors,
(B) 2 transistors and 2 capacitors
(C) 1 transistor and 1 capacitor
(D) 2 capacitors only
Solution: (C)

Sram (Static Ram)

- SRAM is nothing but just a modified form of a latch.
- SRAM cells store the given data like a latch using NOT-gate.

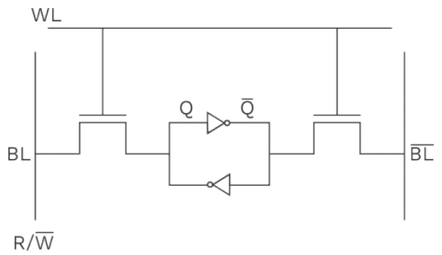


Fig. 7.21

Read operation

- Select the desired cell location using word line (WL) by applying logic '1' of the particular cell. So, the transistors act as short circuits (S.C).
- If Read/Write and word line (WL) is logic '1', then data line (bit line) acts as an output line. So, at the data line (bit line), BL =Q will be stored and $\overline{BL} = \overline{Q}$ will be stored and read operation takes place in SRAM.

Previous Years' Questions

In the DRAM cell in the figure, the V1 of the NMOSFET is 1 V. What could be WL, C and BL voltages when read operation take place.

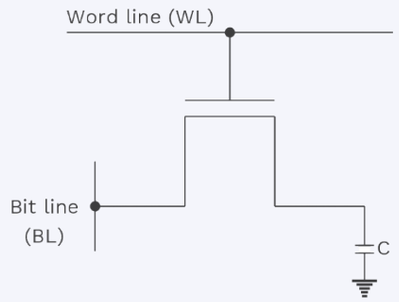


Fig. 7.22

(A) 5 V; 3 V; 7 V
(B) 4 V; 3 V; 4 V
(C) 5 V; 5 V; 5 V
(D) 4 V; 4 V; 4 V
Solution: (B)

Write operation

- Select the desired cell location using word line (WL) by applying logic '1' of particular cell. So, the transistor act as a short circuit (S.C).
- If Read/Write is logic '0' and word line (WL) is logic '1', then data line (bit line) act as an input line. So, at the data line (bit line), BL = Q will be stored and $\overline{BL} = \overline{Q}$ will be stored and write operation takes place in SRAM.

SRAM implementation using CMOS inverter (6 transistor implementation)

In the CMOS inverter, replace the NOT gate of the SRAM circuit with the CMOS circuit.

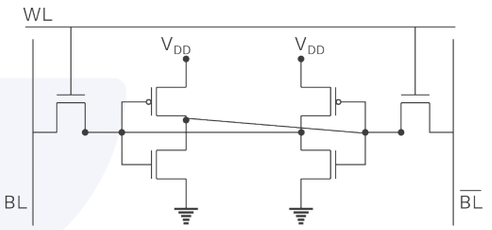


Fig. 7.23

SRAM implementation (using) 4 transistor, replace PMOS with resistor (R)

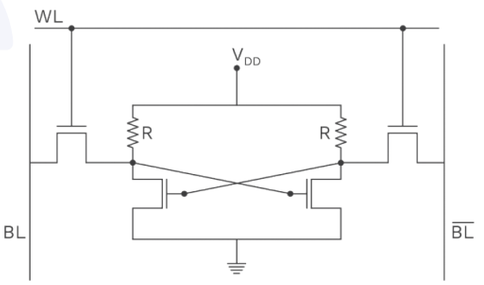


Fig. 7.24

Previous Years' Questions

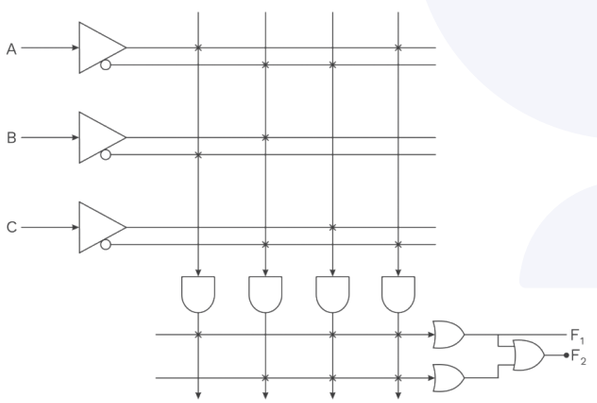
Each cell of a static random access memory contains:
(A) 6 MOS transistors
(B) 4 MOS transistors and 2 capacitors
(C) 2 MOS transistors and 4 capacitors
(D) 1 MOS transistor and 1 capacitors
Solution: (A)

Chapter Summary

- ROM contains a decoder, AND gates and OR gates.
- Semiconductor memories are volatile memories.
- Magnetic disk type memories are non-volatile memories.
- In PROM, AND array is fixed and OR array is programmable.
- In PLA, both AND and OR array are programmable.
- In PAL, OR array is fixed but AND array is programmable.
- In EPROM, the data is stored in floating gates.
- EPROM is also called UV-ROM.
- EEPROM uses electrical signal to erase the data.
- EEPROM is also known as EAROM or FLASH-ROM.
- In SRAM, data is stored in flip-flop like structure.
- In DRAM, data is stored in capacitor.

Practice Questions

1. Find the minterms (F1 and F2) of the given circuit,



(A) F1 = Sm (0, 2, 7), F2 = πM(2)

(B) F1 = Sm (0, 1, 3, 5, 6), F2 = Sm(2, 6)

(C) F1 = Sm (1, 3, 4, 5, 6), F2 = Sm(1, 2, 3, 4, 5, 6)

(D) F1 = Sm (0, 3, 4), F2 = Sm(0, 1, 2, 3, 4)

2. Which of the followings statement (s) is/ are correct regarding the semiconductor memory unit.

(A) PLA contains programmable AND and programmable OR array.

(B) PAL contains programmable OR and programmable AND array.

(C) PAL contains programmable AND and fixed OR array.

(D) PAL contains fixed OR and programmable AND array.

3. If a memory has 12 address lines, then find the number of storage locations available in the memory.

(A) 144

(B) 2048

(C) 1024

(D) 4096

4. Which of the following memories uses a MOSFET and a capacitor as its memory cell?

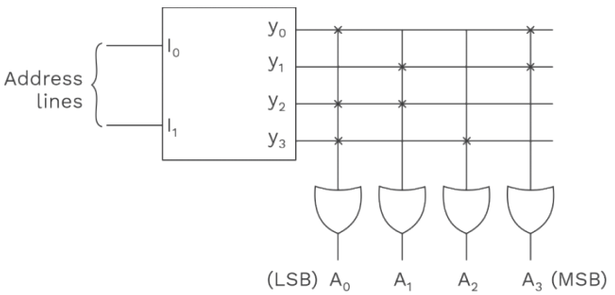
(A) DROM

(B) SRAM

(C) DRAM

(D) ROM

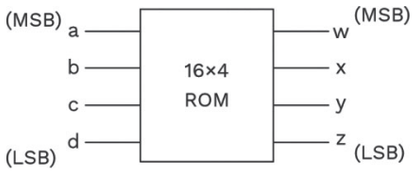
5. The digital circuit shown below is a programmed ROM (designed using 2×4 decoder), where cross (X) shows the connection. The content of PROM at memory location 01 will bebe_____.



(A) 0101
(B) 1011
(C) 1010
(D) 1001

6. How many 16K × 1 RAM chips are needed for the memory capacity of 128K-byte?
(A) 64
(B) 32
(C) 128
(D) 8

7. Specify the decimal equivalent of memory content at address 6 in the following figure:



The relation between input and output are follows:

w(a, b, c, d) = Sm(0, 3, 6, 7, 12)

x(a, b, c, d) = Sm(4, 5, 9, 11, 14)

y(a, b, c, d) = Sm(1, 2, 6, 12, 13)

z(a, b, c, d) = Sm(6, 9, 11, 12)

(A) 8
(B) 11
(C) 9
(D) 7

Practice Solutions

1. Solution: (C) **2. Solution: (A) and (D)** **3. Solution: (D)** **4. Solution: (C)**

5. Solution: (C) **6. Solution: (A)** **7. Solution: (B)**